I Limits of sequences

Definition

A sequence (u_n) is **convergent** to L if $\lim_{n\to\infty} u_n = L$. It is **divergent** if this limit does not exist or equals $\pm\infty$.

II Limits of simple functions

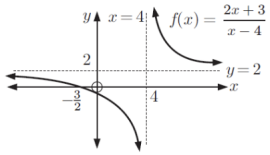
If $f(x)$ can be made as close as we like to some real number A by making x sufficiently close to (but not equal to) a, then we say that $f(x)$ has a **limit** of A as x approaches a, and we write

$$\lim_{x\to a} f(x) = A.$$

In this case, $f(x)$ is said to *converge* to A as x approaches a.
Consider the function

$$f(x) = \frac{2x+3}{x-4}$$

which has domain $\{x \mid x \neq 4,\ x \in \mathbb{R}\}$. From the graph of $y = f(x)$, we can see the function has a vertical asymptote $x = 4$, and a horizontal asymptote $y = 2$.



Both of these asymptotes can be described in terms of limits :

$$\text{As } x \to 4^-, \quad f(x) \to -\infty$$
$$\text{As } x \to 4^+, \quad f(x) \to +\infty$$
$$\text{As } x \to -\infty, \quad f(x) \to 2^-$$
$$\text{As } x \to +\infty, \quad f(x) \to 2^+$$

Since $f(x)$ converges to a finite value as $x \to -\infty$, we write

$$\lim_{x\to-\infty} f(x) = 2.$$

Since $f(x)$ converges to a finite value as $x \to +\infty$, we write

$$\lim_{x \to +\infty} f(x) = 2.$$

Definition

A **vertical asymptote** is a line $x = a$ such that $\lim_{x \to a^{\pm}} f(x) = \pm\infty$. A **horizontal asymptote** is a line $y = L$ such that $\lim_{x \to \pm\infty} f(x) = L$.

III Derivative function

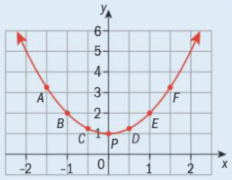
A speed camera takes a single photograph of a car and reports its speed at that exact moment — say 87 km/h. But speed is a *rate of change over time*, and in a single frozen instant nothing moves and nothing changes.

So how can one frozen instant possibly contain a rate of change ? What would it even mean to talk about "speed" at a single point in time ?

What if we tried to estimate the car's speed using two photographs taken very close together in time ? What happens to that estimate as the time gap between the photographs gets smaller and smaller ?

Investigation – secant and tangent lines

Here is the graph of $f(x) = x^2 + 1$



1 Copy the graph to paper and draw lines *AP*, *BP*, *CP*, *DP*, *EP* and *FP*. These lines are called **secant lines** to the graph of $f(x) = x^2 + 1$.

2 Copy and complete the table.

Point	Coordinates	Line	Gradient or slope
P		—	—
A		*AP*	
B		*BP*	
C		*CP*	
D		*DP*	
E		*EP*	
F		*FP*	

Recall that the gradient of a line through the points (x_1, y_1) and (x_2, y_2) is $\frac{y_2 - y_1}{x_2 - x_1}$

3 As points on the curve get closer and closer to point *P*, what value does the gradient of the secant lines seem to be approaching?

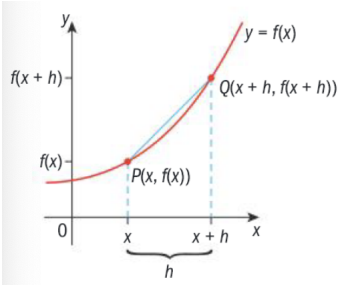
4 Draw the line at point *P* that has the gradient you found in question 3. This line is called the **tangent line** to the graph of $f(x) = x^2 + 1$ at *P*.

Definition

Given two points $P(x,f(x))$ and $Q(x+h,f(x+h))$ on the curve $y=f(x)$, the **gradient of the secant line** PQ is

$$\frac{f(x+h)-f(x)}{(x+h)-x}=\frac{f(x+h)-f(x)}{h}.$$

The expression $\frac{f(x+h)-f(x)}{h}$ is called the **difference quotient**.



Example

Write an expression for the gradient of a secant line to the curve $f(x)=x^2+1$ and simplify it.

Suppose that point Q slides down the curve and approaches point P. The secant lines PQ will get closer to the tangent line at point P. As Q gets closer to P, h gets closer to 0. We can take the limit as h approaches 0 of the gradient of the secant line to get the gradient of the tangent line :

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}.$$

This limit is not a constant. It is a function that gives the gradient of f at x.

Definition [HL only]

The function $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ is the **derivative** of f.

Reading the notations. $f'(x)$ is read "the derivative of f", "f prime of x", or $\frac{dy}{dx}$ "the derivative of y with respect to x".

Remark. When we evaluate this limit to find a derivative function, we say we are **differentiating from first principles**

Gradient of the tangent at $x = a$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

Example

Find the derivative of $f(x) = x^2 + 1$ and hence find the gradient of the tangent line when $x = 3$.

IV Differentiation rules

a. Basic rules

Basic differentiation rules

For differentiable functions u,v and a constant c :

$f(x)$	Name of rule	$f'(x)$
c (a constant)	differentiating a constant	0
x^n	differentiating x^n	nx^{n-1}
$cu(x)$	constant times a function	$cu'(x)$
$u(x) + v(x)$	addition rule	$u'(x) + v'(x)$

b. Product rule

Investigation

Suppose $u(x)$ and $v(x)$ are two functions of x, and that

$$f(x) = u(x)v(x)$$

is the product of these functions. The purpose of this investigation is to find a rule for determining $f'(x)$. What to do :

1. Suppose $u(x) = x$ and $v(x) = x$, so $f(x) = x^2$.
   (a) Find $f'(x)$ by direct differentiation.
   (b) Find $u'(x)$ and $v'(x)$.
   (c) Does $f'(x) = u'(x)v(x)$?
2. Suppose $u(x) = x$ and $v(x) = \sqrt{x}$, so $f(x) = x\sqrt{x} = x^{3/2}$.
   (a) Find $f'(x)$ by direct differentiation.
   (b) Find $u'(x)$ and $v'(x)$.
   (c) Does $f'(x) = u'(x)v(x)$?

3. Copy and complete the following table, finding $f'(x)$ by direct differentiation.

$f(x)$	$f'(x)$	$u(x)$	$v(x)$	$u'(x)$	$v'(x)$	$u'(x)v(x) + u(x)v'(x)$
x^2		x	x			
$x^{3/2}$		x	$\sqrt{x}$			
$x(x+1)$		x	$x+1$			
$(x-1)(2-x^2)$		$x-1$	$2-x^2$			

4. Copy and complete : *"If $u(x)$ and $v(x)$ are two functions of x and $f(x) = u(x)v(x)$, then $f'(x) = \ldots$ "*

Theorem

Product rule

If $u(x)$ and $v(x)$ are two functions of x and $f(x) = u(x)v(x)$, then

$$f'(x) = u'(x)v(x) + u(x)v'(x).$$

c. Quotient rule

Theorem

If

$$Q(x) = \frac{u(x)}{v(x)} \quad \text{then} \quad Q'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2}.$$

Or if

$$y = \frac{u}{v}, \quad \text{where } u \text{ and } v \text{ are functions of } x, \quad \frac{dy}{dx} = \frac{u'v - uv'}{v^2}.$$

Proof

Quotient functions have the form

$$Q(x) = \frac{u(x)}{v(x)}.$$

Notice that

$$u(x) = Q(x)v(x),$$

and by the product rule

$$u'(x) = Q'(x)v(x) + Q(x)v'(x).$$

$$\therefore u'(x) - Q(x)v'(x) = Q'(x)v(x)$$

$$\therefore Q'(x)v(x) = u'(x) - \frac{u(x)}{v(x)}v'(x)$$

$$\therefore Q'(x)v(x) = \frac{u'(x)v(x) - u(x)v'(x)}{v(x)}$$

$$\therefore Q'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2}, \quad \text{when this exists.}$$

Example

Use the quotient rule to find $\frac{dy}{dx}$ if :

$$y = \frac{1 + 3x}{x^2 + 1}.$$

d. Chain rule

Theorem

If $y = g(u)$ where $u = f(x)$ then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

This rule is extremely important and enables us to differentiate complicated functions much faster.

Property

For any function $f(x)$:

If $y = [f(x)]^n$ then

$$\frac{dy}{dx} = n[f(x)]^{n-1} \times f'(x).$$

Example

Find $\frac{dy}{dx}$ if :

1. $y = (x^2 - 2x)^4$
2. $y = \frac{4}{\sqrt{1-2x}}$

e. Exponential functions

Theorem

If $f(x) = b^x$ then $f'(x) = kb^x$ where k is a constant equal to $f'(0)$.

Proof

[HL ONLY] If $f(x) = b^x$, then

$$f'(x) = \lim_{h\to 0} \frac{b^{x+h} - b^x}{h} \qquad \{\text{first principles definition of the derivative}\}$$

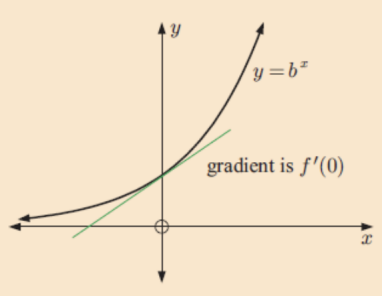
$$= \lim_{h\to 0} \frac{b^x(b^h - 1)}{h}$$

$$= b^x \times \left(\lim_{h\to 0} \frac{b^h - 1}{h}\right) \qquad \{\text{as } b^x \text{ is independent of } h\}$$

But

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{b^h - 1}{h}.$$

$$\therefore f'(x) = b^x f'(0).$$



Given this result, if we can find a value of b such that $f'(0) = 1$, then we will have found *a function which is its own derivative!*

Investigation

Graph $f(x) = b^x$ and its derivative function $y = f'(x)$. Experiment with different values of b until the graphs of $f(x) = b^x$ and $y = f'(x)$ appear the same. Estimate the corresponding value of b to 3 decimal places.

Theorem

If $f(x) = e^x$ then $f'(x) = e^x$.

f. Derivative of $e^{f(x)}$

Function	Derivative
e^x	e^x
$e^{f(x)}$	$e^{f(x)} \cdot f'(x)$

Proof

[HL ONLY] $e^{f(x)} > 0$ for all x, no matter what the function $f(x)$. Suppose

$$y = e^{f(x)} = e^u \quad \text{where } u = f(x).$$

Now

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \{\text{chain rule}\}$$
$$= e^u \cdot \frac{du}{dx}$$
$$= e^{f(x)} \cdot f'(x).$$

g. Derivative of $\ln f(x)$

Property

If $y = \ln x$ then

$$\frac{dy}{dx} = \frac{1}{x}.$$

Function	Derivative
$\ln x$	$\frac{1}{x}$
$\ln f(x)$	$\frac{f'(x)}{f(x)}$

Proof

Suppose $y = \ln f(x)$. So $y = \ln u$ where $u = f(x)$. Now

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \{\text{chain rule}\}$$

$$\frac{dy}{dx} = \frac{1}{u}\frac{du}{dx}$$
$$= \frac{f'(x)}{f(x)}.$$

Example

Find the gradient function of :

$$\text{a) } y = \ln(kx), \quad k \text{ a constant} \qquad \text{b) } y = \ln(1-3x).$$

Remark. Laws of Logarithms For $a > 0,\ b > 0,\ n \in \mathbb{R}$:

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a.$$

Example

Differentiate with respect to x :

$$\text{a) } y = \ln(xe^{-x}), \qquad \text{b) } y = \ln\left(\frac{x^2}{(x+2)(x-3)}\right).$$

h. Second and Higher Derivatives

Given a function $f(x)$, the derivative $f'(x)$ is known as the **first derivative**.

Definition

The **second derivative** of $f(x)$ is the derivative of $f'(x)$, or the derivative of the first derivative. We use $f''(x)$ or y'' or $\dfrac{d^2y}{dx^2}$ to represent the second derivative.

$f''(x)$ reads "f double dashed x".

$\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right)$ reads "dee two y by dee x squared".

Example

Find $f''(x)$ given that

$$f(x) = x^3 - \frac{3}{x}.$$

We can continue to differentiate to obtain higher derivatives.

Definition

The **nth derivative of y with respect to x** is obtained by differentiating $y = f(x)$ n times. We use the notation $f^{(n)}(x)$ or $\dfrac{d^ny}{dx^n}$ for the nth derivative.